TopoMiner: Efficient Topology Discovery for IPv6 Networks

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***Abstract*—Topology discovery can be used to obtain the connectivity and operational status of network devices by discovered interfaces. By performing topology discovery on networks, administrators can better understand and manage networks and improve network reliability and security. However, the large IPv6 address space, sparse address distribution, and unknown address assignment policies make it infeasible to simply and roughly probe the IPv6 network topology.**

To this end, we design an efficient IPv6 interface-level topology discovery method called TopoMiner. It performs multiple rounds of probing the IPv6 topology with the given set of IPv6 prefixes and locates the high-density interface regions based on the results of each round of topology discovery. TopoMiner then performs prefix expansion and address generation for the high-density interface address regions. At the same time, TopoMiner also uses some of the prefixes that have been probed to regenerate the target addresses. In this way, TopoMiner can dynamically adjust the breadth and depth of topology discovery and thus complete multiple rounds of topology discovery within the packet-sending budget. In real-word probing, TopoMiner achieves a 3∼7× enhancement in probing efficiency compared to state-of-the-art topology discovery methods.

Keywords—IPv6, topology discovery, high-density interface address regions, target addresses

I. Introduction

With the escalating issue of IPv4 address exhaustion, the global adoption of IPv6 is gradually increasing [1]. Statistics from the Internet Assigned Numbers Authority (IANA) [2] and the Regional Internet Registry (RIR) reveal a significant number of IPv6 address blocks have been allocated worldwide [3]–[8]. Internet Service Providers (ISPs) actively embrace IPv6 to offer uninterrupted Internet connectivity and communication to their subscribers [9]. Prominent websites and online service providers like Google [3] and Facebook [4] have taken the initiative to enable IPv6 support, ensuring that users can access their platforms and services seamlessly via IPv6. These collective efforts play a crucial role in propelling the development of IPv6.

The development of IPv6 networks necessitates comprehensive and in-depth network topology discovery, which is vital in optimizing network performance [10], identifying network structures [11], diagnosing network failures [12], and ensuring real-time network security supervision [13]–[15]. However, the large and sparse IPv6 address space, coupled with unknown address allocation policies, poses a significant challenge for network topology discovery.

Despite existing efforts to optimize IPv4 topology discovery methods [16], [17], they do not address the challenges faced in topology discovery under IPv6. As a result, several studies have attempted to optimize topology probing targets to improve efficiency. Robert et al. [5] conduct IPv6 topology discovery by performing prefix transformation and target address synthesis based on multiple complementary seed sets that have been selected. However, this strategy only probes within an initially given prefix set and lacks dynamic adaptation of the probing direction based on the probing results, resulting in limited topology discovery scope and efficiency. Subsequently, 6Search [18] utilizes reinforcement learning to automatically adjust resource allocation based on probing results to accomplish IPv6 topology discovery while improving probing efficiency[1]. However, this approach suffers from premature convergence, leading to a reduction in the number of probed interfaces.

To this end, we propose TopoMiner, a novel topology discovery method that can discover more interface addresses with limited probing resources. The central concept of TopoMiner lies in the continuous and extensive exploration of high-density router interface regions, coupled with the ongoing allocation of probing resources to these regions. This enables a persistent topology discovery process towards the target region. However, it is not trivial to realize the above idea, and we mainly face the following challenges: 1) The exact distribution of high-density router interface regions is unknown, posing a major challenge in locating them with limited probing resources. Random probing, without prior identification of these regions, can result in inefficiencies. 2) Probing often becomes concentrated in localized regions, leading to a significant investment of probing resources but resulting in the discovery of only a small number of interfaces. Therefore, TopoMiner incorporates the following design. First, TopoMiner employs the concept of multiple rounds of probing and subsequently extracts common prefixes from the interfaces obtained in each round of probing to determine the direction of the next round of probing. This methodology not only aids

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[1]The probing efficiency is the ratio of the number of addresses discovered to the number of packets sent.

in identifying high-density interface regions but also enables comprehensive topology discovery within these targeted regions. Second, TopoMiner expands on common prefixes and also employs a weighted random sampling technique to select previously detected prefixes and regenerate target addresses. This approach allows for fine-tuning the breadth of topology discovery while ensuring a relatively uniform distribution of probe resources across the target area. We deploy TopoMiner in practice by probing both BGP-announced routing prefixes and the set of /48 prefixes associated with active addresses in a publicly available hitlist [19]. Our results demonstrate improved performance compared to state-of-the-art approaches.

Contributions. The main contributions of this paper are:

- We propose an IPv6 topology probing method, TopoMiner, which accurately locates high-density interface regions and continuously explores them in-depth. Additionally, it evenly allocates limited probing resources to all target regions where high-density interface regions are discovered, enabling a more comprehensive topological discovery of the target regions.
- We implement a prototype of TopoMiner and conduct real-world probing, achieving a 3$\sim$7$\times$ enhancement in topology discovery efficiency compared to existing state-of-the-art probing methods.
- For the reproducibility of our work, we also make the code and data publicly available at https://github.com/sshwei/TopoMiner.git.

The rest of the paper is organized as follows. In §II, we briefly review various existing topology discovery tools and methods. §III gives a high-level overview of TopoMiner. In §IV, we describe each key point of TopoMiner in detail. In §V, we evaluate the performance of TopoMiner. Finally, we conclude the paper in §VI.

II. Related Work

In IPv4 network topology discovery, researchers have made significant efforts and developed various topology discovery tools [11]. One commonly used tool is `traceroute`, which provides accurate path information at the interface level. It utilizes the Time Exceeded message of the Internet Control Message Protocol (ICMP) for probing [20]. When a packet passes through a router, the router decreases the Time to Live (TTL) value of the packet and checks if it reaches zero. If the TTL value becomes zero, the router drops the packet and sends a Time Exceeded message to the source host. By iterating this process, `traceroute` can determine the path of the packet from the source host to the destination host. Fig. 1 shows the entire process of topology discovery using `traceroute` from the source host to the target host 2001:252:0:2::101. However, `traceroute` has limitations when it comes to load balancing, as it cannot accurately trace a single path. Moreover, when used for discovering topology with multiple target addresses, `traceroute` can lead to redundant probing. Additionally, its slow probing rate also makes it unsuitable for large-scale IPv6 topology discovery. To address these issues, the research community has developed improved topology discovery tools. Paris Traceroute [16] is one such tool designed to provide more accurate measurements of network paths. It achieves this by controlling the contents of the packet header, thereby overcoming the challenges posed by load balancing. The Multicast Dissemination Algorithm (MDA) [21] discovers all possible paths of a load-balanced path by sending a large number of probe packets. MDA-Lite [22] is an enhanced version of MDA that reduces the overhead of topology discovery by leveraging common routing patterns and "diamond uniformity" features while maintaining a low probability of failure. Doubletree [23] utilizes the tree structure of routing to aggregate paths from a single source to multiple destinations or from multiple sources to a single destination, effectively reducing redundant probes. Another topology discovery tool, Scamper [24], supports both IPv4 and IPv6. It provides basic path probing features and flexible plug-in mechanisms, facilitating the implementation of Doubletree and Paris Traceroute. Scamper also implements four techniques to infer whether an observed IP address in the Internet topology is an alias.

Yarrp6 [17] revolutionizes the traditional probing model by eliminating the need to maintain additional state information. This innovation is achieved by embedding all necessary states into the probe, enabling the received ICMPv6 TTL exceeded message to contain information about the probe. Yarrp6 enhances probing speed by decoupling the send and receive packet processes and employs random-hop probing to circumvent ICMPv6 rate-limiting [12]. It also adapts to load-balanced paths by maintaining fixed values for specific fields and selecting appropriate probing modes. This adaptation is made possible by parsing and computing relevant probing information from received ICMP TTL Exceeded messages. Furthermore, FlashRoute [25] is a topology discovery tool that combines the separated sending and receiving packet processes from Yarrp6 with the adaptive probing strategy of Doubletree. This composition significantly improves topology discovery efficiency. Each of these advancements in topology discovery tools addresses specific limitations and contributes to improving the efficiency and accuracy of network topology discovery.

To achieve a more comprehensive and in-depth exploration during topology discovery, it is crucial to utilize reliable probing methods in addition to probing tools [26]. Current topology discovery methods include Robert Beverly's approach, which involves generating a target address set using multiple prefix sets and conducting one-time topology discovery on these target addresses using Yarrp6 to obtain relevant topology information [5]. However, such methods often suffer from limited topology coverage within a given set of prefixes. To address this limitation, a novel approach called 6Search has been proposed. It focuses on topology discovery for addresses generated under the /32 prefixes [7], [18]. This method incorporates a reinforcement learning-based address generation technique, enabling dynamic adjustment of probing resource allocation. It allocates more probing resources to high-value prefixes while assigning fewer resources to low-

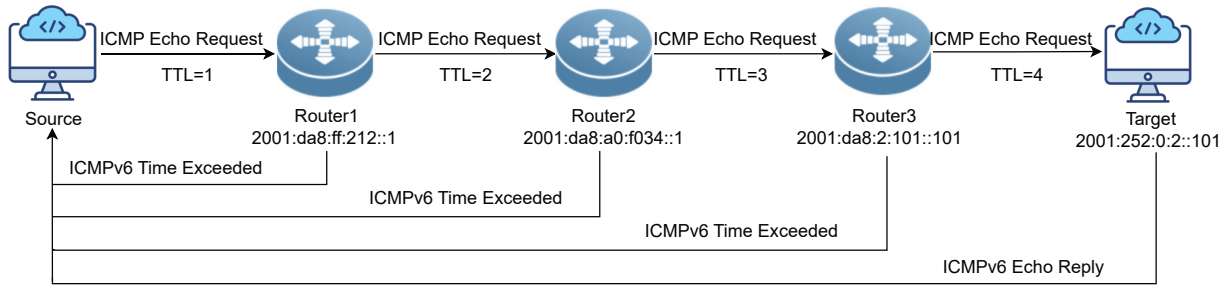

Fig. 1. The process of topology discovery with traceroute.

value prefixes. Additionally, 6Search leverages the Doubletree algorithm to minimize redundant probes. However, this method has limitations in terms of both the breadth and depth of topology discovery, which prevents it from achieving a more comprehensive and in-depth exploration of the network topology.

Overall, these advancements in topology discovery methods aim to enhance the comprehensiveness and depth of network topology exploration. However, they still face certain challenges that need to be addressed to achieve more comprehensive and in-depth results.

III. TopoMiner Overview

In this section, we present a high-level overview of TopoMiner including the architecture and workflow. The goal of TopoMiner is to automatically adjust the breadth and depth of topology discovery, enabling a more comprehensive and in-depth exploration of topology. Fig. 2 illustrates the high-level overview of topology discovery undertaken by TopoMiner, which involves several functional modules:

- **Prefix Determination**: This module is used to identify the prefixes for performing topology discovery. For the first round of probing, the routing prefixes announced by BGP are used directly. For subsequent rounds, subprefixes with high yield are selected from the generated common prefixes for probing.
- **Target Generation**: This module is responsible for generating the target addresses under the prefixes to be probed.
- **Target Probing**: It probes the network path from the vantage point to the target addresses. The obtained traces can be utilized to identify regions with dense interface addresses.
- **Common Prefix Extraction**: This module is used to obtain common prefixes between neighboring hops in the trace to facilitate further probing of regions with dense interface addresses.

Workflow. Initially, TopoMiner utilizes the prefix determination module to identify the target prefixes to be probed. Subsequently, the target generation module generates target addresses for probing. Upon completing the probing phase, TopoMiner utilizes the results from each round to extract common prefixes associated with the discovered interface addresses. Moreover, TopoMiner identifies regions with a high density of interface addresses and extends the extracted common prefixes to encompass these regions, allowing for deeper exploration of the provided prefix set. To enhance the comprehensiveness of topology probing, TopoMiner randomly assigns weights to some of the previously probed prefixes. Consequently, the newly generated prefixes are merged with the old ones to create a refreshed prefix set. Subsequently, TopoMiner employs this refreshed set of prefixes to generate new target addresses for the subsequent round of topology discovery. Upon reaching a predetermined budgeted number of sent packets for topology discovery, TopoMiner concludes the probing process. This adaptive approach allows TopoMiner to dynamically adjust the depth and breadth of the exploration process, resulting in a more comprehensive discovery of the topology associated with a given prefix set.

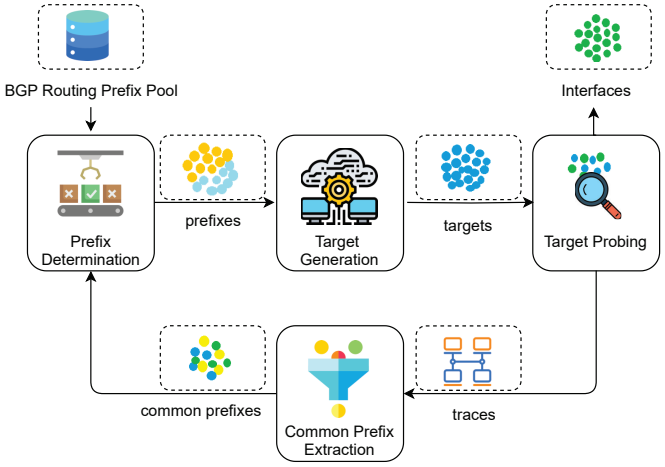

Fig. 2. High-level overview of TopoMiner.

IV. Design Details

TopoMiner is an efficient topology discovery method that dynamically adjusts the breadth and depth of topology exploration based on the probing results. It extracts prefixes and performs relevant operations on existing prefixes to achieve a more comprehensive and in-depth topology discovery on a limited budget. In this section, we present the details of TopoMiner for topology discovery, mainly including 1) prefix determination (§IV-A), 2) target generation and probing (§IV-B), and 3) common prefix extraction (§IV-C).

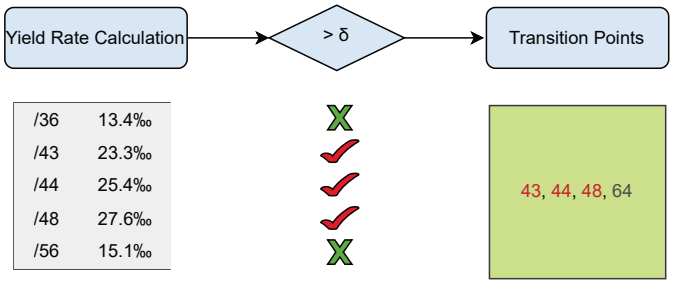

Fig. 3. Determination of transition points ($\delta = 20‰$). The ✓ in the figure indicates the yield corresponding to a prefix length is greater than the threshold, while the X indicates the opposite. The prefix length that satisfies the threshold will be selected as the transition point.

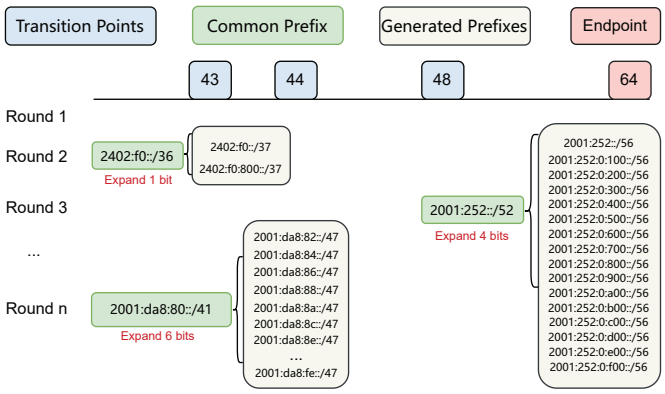

Fig. 4. Generation of prefixes to be probed based on common prefixes.

A. Prefix Determination

The significance of prefix determination is to identify the prefixes to be probed that potentially have rich router interfaces, which in turn determines the probing efficiency of TopoMiner. It consists of two essential components: prefix expansion and prefix selection. On the one hand, prefix expansion allows for a more in-depth exploration of the established prefix set in topology discovery. On the other hand, prefix selection extends the breadth of topology discovery. The combination of these two components enables a more comprehensive and in-depth topology discovery. The following is a detailed explanation of both components.

1) Prefix Expansion: The goal of prefix expansion is to expand the extracted common prefixes to encompass as many sub-prefixes as possible covering interface addresses that can be probed, facilitating the acquisition of a more comprehensive topology. We detail common prefix extraction in §IV-C. Due to limited probing resources, enumerating all sub-prefixes under a common prefix for probing is impractical. Therefore, TopoMiner selects only some sub-prefixes under each common prefix for probing. Fig. 3 and Fig. 4 show the concrete process of prefix expansion in TopoMiner, which employs a two-step prefix expansion strategy. Note that the input for the first round of probing is the routing prefixes announced by BGP without having to make additional expansions.

Firstly, TopoMiner defines multiple transition points for prefix expansion length. These transition points represent the signs of entering high-density interface regions. Each common prefix should be extended as far as possible to the nearest transition point, allowing it to cross over the transition point and penetrate deeper into the high-density interface region. Therefore, setting these transition points effectively is crucial. Fig. 3 shows how TopoMiner dynamically acquires the transition points used for each round of probing. Initially, TopoMiner establishes endpoint for prefix expansion. Based on RFC4291 [27], the length of an interface identifier (IID) is fixed at 64 bits. Consequently, the expanded prefix length is capped at 64 bits. Additionally, the transition points below level 64 will be selected during each round of probing. The selection of these transition points is contingent upon the outcomes of each round of probing. TopoMiner calculates the yield rate of interfaces for each prefix class based on the results of the current round. The interface yield is defined as the ratio of the incremental number of interfaces discovered when probing a specific address class in a round of topology probing to the incremental number of packets sent when probing that address class in the same round. We classify the addresses corresponding to the prefix to be probed based on the prefix length. A high interface yield rate indicates a dense concentration of newly discovered interface addresses within that prefix class. We establish a threshold δ with a value between 0 and 1, and any prefix class with an interface yield rate greater than this threshold is deemed high. The lengths of these high-yield prefix classes serve as the transition points for our selection process. As shown in Fig. 3, we take the value of δ to be 20‰, and then the three expansion transition points identified are 43, 44, and 48.

Secondly, TopoMiner determines the basic number of bits (g) to be expanded for each common prefix. The typical operation of TopoMiner involves extracting common prefixes after each round of probing to determine the targets for the subsequent round. Upon close examination, we observe that the initial extraction of common prefixes from the probing results often yields a high number in the first round. To prevent the wastage of probing resources through excessive expansion, we opt for a conservative expansion approach for these prefixes by setting g to 1 in the first round. In subsequent rounds, where the number of extracted prefixes is typically smaller, we set g to 4 to enable deeper probing. After determining the basic number of expansion bits, when the distance between the common prefix and its nearest transition point exceeds g, TopoMiner generates 2^g sub-prefixes under the common prefix for topology probing. This approach can prevent excessive expansion caused by the distance to the nearest transition point being too far. As shown in Fig. 4, the common prefixes 2402:f0::/36 and 2001:252::/52 are expanded by 1 and 4 bits in the second and third rounds, respectively, generating 2 and 16 prefixes to be probed in the next round. Note that if the difference between the length of a common prefix (l) and its nearest transition point (n) is less than or equal to g, then $n-l+g$ bits need to be extended, resulting in the generation of 2^{n-l+g} sub-prefixes. For example, the common prefix 2001:da8:80::/41 in

Fig. 4 is expanded by 6 bits. This way it can be expanded through transition points to /47, generating 64 /47 prefixes to be used in the next round of probing. TopoMiner then de-duplicates these newly generated prefixes in preparation for the next round of topology discovery.

2) Prefix Selection: The purpose of this step is to prevent the topology discovery of TopoMiner from converging prematurely and being limited to localized regions. To achieve the above goal, TopoMiner first maintains a prefix pool that contains the probed prefixes. Then, TopoMiner not only expands from the common prefixes to get the target prefixes to be probed in the next round but also may extract the previously probed prefixes from the prefix pool for probing.

The details of the approach are as follows. TopoMiner is directly using all BGP-announced routing prefixes for probing in the first round. Assume that in each round after the second round, the probing budget is b (initial value is 0) and the number of target prefixes obtained by prefix expansion is m. If m is less than b, in addition to selecting these m target prefixes, $b-m$ prefixes from the prefix pool are selected for probing in the next round; if m is greater than or equal to b, these m target prefixes are directly probed in the next round while updating the value of b to m. As for the method of selecting the $b-m$ prefixes from the prefix pool, we use weighted random sampling. As mentioned earlier, we categorize the prefixes by length and then sample according to the percentage of prefixes in each class. Random sampling is then used in each class of prefixes. This approach allows us to effectively prevent TopoMiner from using a large budget to detect only a small range of prefixes, expanding the scope of probing and reducing wasted resources.

B. Target Generation & Probing

To generate the target addresses, we synthesize the prefixes selected in the previous step with interface identifiers (IIDs) [27]. There are various methods to generate IIDs, such as using ::1 addresses or randomized ones [28]. Typically, ::1 addresses are assigned to critical devices like routers [28]. However, to ensure non-intrusive probing, we opt for randomized interface identifiers to synthesize target addresses.

For each prefix, TopoMiner synthesizes an address accordingly. After synthesizing the set of target addresses, we perform de-duplication to ensure uniqueness. Subsequently, we conduct topology discovery on these target addresses, obtaining traces that reach the target addresses from our vantage point. It is worth mentioning that if there is a "*" hop in the trace, we ignore it.

C. Common Prefix Extraction

The objective of this step is to extract the longest common prefix from the interface addresses of neighboring hops obtained in each round of probing. This facilitates the identification of regions with high-density interface addresses, allowing for targeted probing to enhance efficiency.

Fig. 5 illustrates in detail the extraction process of common prefixes. First, TopoMiner groups the interface addresses in each trace from the current round of probing results by the autonomous system (AS). Typically, probes sent out by the vantage point will traverse multiple autonomous systems to reach their destinations. Therefore, we can determine the AS to which each interface address in each trace belongs, and then categorize the interface addresses according to the ASes. This categorization helps to evaluate the density of interface addresses found in different prefixes and provides a basis for adjusting the probing resources and directions. In the described process, we may encounter addresses for which we cannot determine the AS to which they belong, such as the address 2001:7fa:0:1::ca28:a0f6 in Fig. 5. These instances occur in situations where: 1) the prefix to which they belong is not announced in the BGP, or 2) the list of routing prefixes announced by BGP, which we utilize, omits the routing prefix to which the address belongs. In either case, we exclude these addresses from further consideration. TopoMiner then checks whether there are cases in which any of these groups contain only a single interface address. If so, TopoMiner directly extracts its routing prefix for subsequent probing. For example, 2400:cb00:581:3:: in the figure belongs to the above case, and we will extract its routing prefix 2400:cb00:581::/48 for subsequent probing. Then, TopoMiner adopts the approach of directly extracting the common prefix between each pair of neighboring interface addresses in the group for subsequent deeper probing. This is based on the observation that the similarity of addresses of neighboring hops suggests a higher likelihood of belonging to the same common prefix. Additionally, TopoMiner utilizes a binary search algorithm for extracting the longest common prefix. Considering the actual allocation of IPv6 addresses and the composition of IPv6 addresses, the lengths of common prefixes extracted are different. We only choose common prefixes with lengths between 16 and 64. After extracting the common prefixes for all groups, TopoMiner conducts a de-duplication operation on these prefixes. The de-duplicated list of prefixes is then used as input for the next stage of prefix determination.

V. Evaluation

We have implemented a prototype of TopoMiner and evaluated its performance. This section shows the evaluation results. All probing experiments are performed on an Ubuntu 20.04 server with an eight-core Intel Core Processor (Broadwell)@2.19GHz and 32 GB of RAM. TopoMiner employs Yarrp6 to probe each target address. To mitigate network warnings, we configure Yarrp6 with a probing rate of 5Kpps, a TTL of 16, and utilize ICMPv6 probes. Additionally, we enable the "-n" option in Yarrp6 to minimize redundant probing. We set the threshold δ for high-density interface regions to "20‰".

A. Experimental Setup

Datasets. As presented in Table I, we utilize two datasets for evaluation:

- **BGP-0215**: This dataset contains global routing prefixes announced by BGP, obtained from the RouteViews

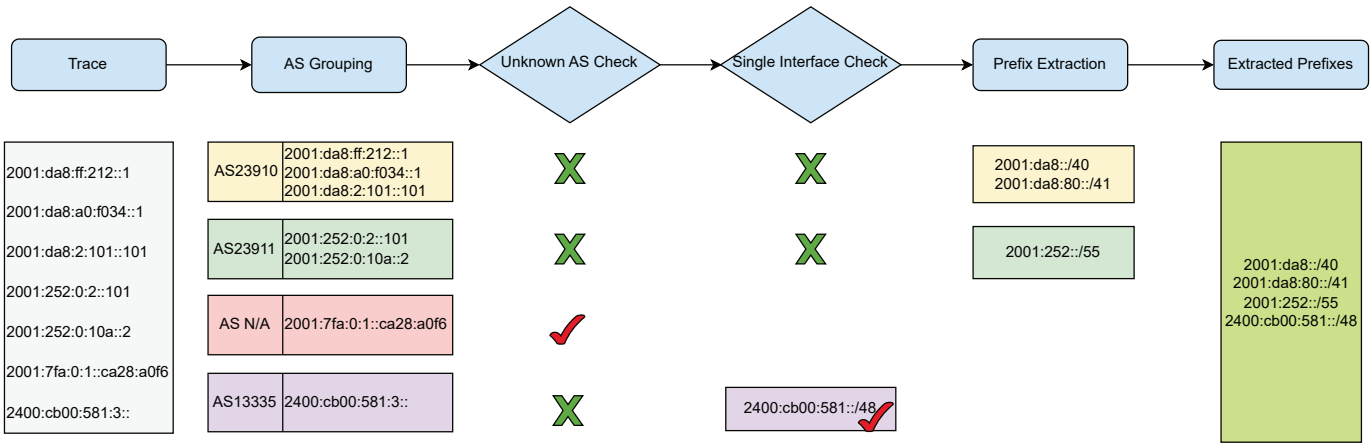

Fig. 5. The extraction process of common prefixes. The interface addresses are extracted from the traces, then grouped by AS, and finally, the common prefix of the neighboring interface addresses within each group is extracted. The ✓ in the figure indicates compliance with the current check option, while the X indicates the opposite.

TABLE I
DATASETS

Name	#Prefixes	#ASNs	IID Type	Date
Hitlist-0215	206,244(11.49%)	10,181(31.23%)	Randomized	2024/02/15
BGP-0215	206,244(96.57%)	32,585(99.95%)	Randomized	2024/02/15

project [29]. This project, maintained by multiple network operators, collects and archives BGP data globally, including IPv6 routing information. We use data published by this program on February 15, 2024.

- **Hitlist-0215**: This dataset includes the /48 prefixes to which addresses in a well-known hitlist belong. The hitlist is provided by a publicly available service [30] and contains a large number of responsive IPv6 addresses. To form a new set of hitlist prefixes, we extract and deduplicate the /48 prefixes of over 13 million responsive addresses provided by the hitlist published on February 15, 2024.

The coverage of BGP-0215 in these two datasets is more extensive, while the prefixes in Hitlist-0215 are more specific. Using both at the same time allows for more comprehensive and in-depth topology discovery. For target generation, we generate a pseudo-random interface identifier for each prefix in both datasets separately, combining them to form a random IPv6 address.

Baselines. We compare TopoMiner with two state-of-the-art approaches, 6Search [18] and Yarrp6 [5]. 6Search utilizes a reinforcement learning approach to generate addresses and dynamically allocate topology probing resources for topology discovery. Yarrp6 does not generate new addresses during topology discovery and relies on the original target address for the discovery process. All three methods are compared with the same budget (i.e., total number of probes).

Evaluation Metrics. We mainly consider the following evaluation metrics:

- **Efficiency**: It is the ratio between the total number of discovered interface addresses and the total number of packets sent.
- **Breadth**: It involves using the BGP routing prefix set to query the number of ASes and the number of routing prefixes associated with the discovered addresses. It can be used to measure the breadth of topology discovery.
- **Depth**: It is the number of discovered addresses matched to under each routing prefix. We use it to measure the depth of topology discovery.

B. Performance

1) Probing Efficiency: We conduct topology discovery using TopoMiner and two baselines under probing budgets ranging from 10M to 50M. The probing results for datasets BGP-0215 and Hitlist-0215 are shown in Fig. 6 and Fig. 7, respectively. In order to make a fair comparison, we will remove redundant aliased addresses from the discovered interface addresses based on the aliased prefix list [30], and only keep one interface address under each aliased prefix. It is evident that TopoMiner outperforms both baselines on both datasets. At a budget of 50M, TopoMiner discovers a much higher number of addresses than the two baselines, showing an improvement of 3-7 times; the probing efficiency is also enhanced by 3-7 times compared to the two baselines. This is primarily due to TopoMiner's ability to locate high-density router interface address regions based on the results of each round of probing and then conduct further probing in those regions.

Both 6Search and TopoMiner employ address regeneration for topology discovery, but TopoMiner demonstrates significantly higher efficiency in its probing. Further, we thoroughly analyze the distribution of addresses discovered by TopoMiner and 6Search. We select the top 25 prefixes with the highest number of addresses found by each method to compare the number of addresses found by the other method with the same prefixes. The results are shown in Fig. 8 and Fig. 9, respectively. Among the 25 prefixes in which 6Search discovers the most interface addresses, TopoMiner

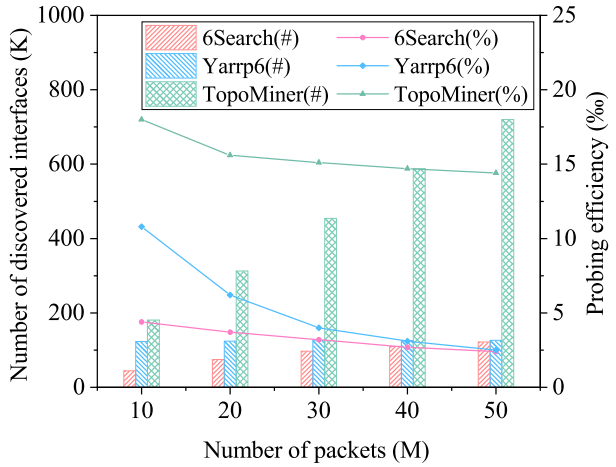

Fig. 6. Comparison of discovered addresses and probing efficiency under the dataset BGP-0215.

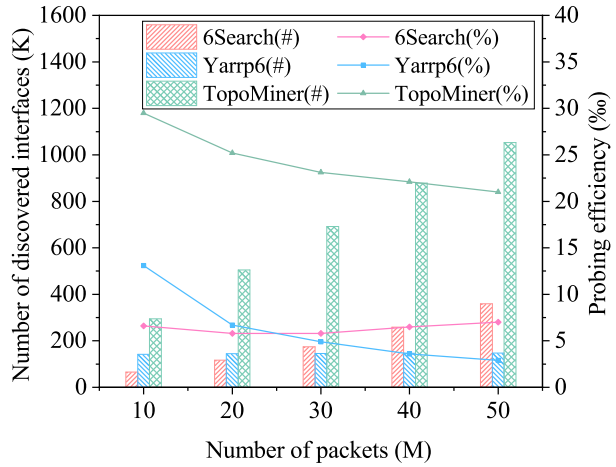

Fig. 7. Comparison of discovered addresses and probing efficiency under the dataset Hitlist-0215.

TABLE II
COMPARISON OF BREADTH

Method	#Interfaces Discovered	#ASes	#Routing Prefixes
6Search	122,081	8,287	26,294
Yarrp6	125,955	6,847	16,897
TopoMiner	**719,722**	**9,776**	**30,487**

discovers more interface addresses than it does in the vast majority of prefixes. However, among the 25 prefixes where TopoMiner discovers the most interface addresses, 6Search finds no interface addresses in the majority of these prefixes. This strongly suggests that TopoMiner's strategy of finding high-density interface address regions based on the interface addresses discovered in each round of probing is effective in discovering new interface addresses and improves the probing efficiency.

2) Breadth: We compare the three methods in terms of the number of ASes and the number of BGP routing prefixes to which the interface addresses discovered by them belong. The results are presented in Table II. TopoMiner outperforms the two baselines in both of these aspects. This means that TopoMiner is more capable of discovering a wider range of network topologies during the discovery process.

Based on the experimental results of topology discovery on the BGP address set, we conduct a comparison of the number of ASes and the number of BGP routing prefixes among the three methods. The breadth metrics are calculated using Pyasn [31] based on the same data as presented in table I. The results in table II clearly demonstrate that TopoMiner outperforms the other two methods in terms of both the number of ASes and the number of BGP routing prefixes. This implies that TopoMiner has a greater ability to uncover a more extensive range of network topologies during the discovery process.

3) Depth: We also compare the depth of topology discovery by the three methods. We first categorize the interface addresses discovered by the three methods according to the length of the BGP-announced routing prefixes to which they belonged. Then, we calculate the average number of interface addresses per prefix category based on the number of matching routing prefixes. Fig. 10 shows the counts and calculations. It is evident that TopoMiner discovers a much higher number of addresses compared to the other two baselines. This indicates that TopoMiner is capable of deeper topology discovery compared to the other two baselines.

C. Module Effects

In addition, we have evaluated the effect of the design of the different parts of the TopoMiner. For this purpose, We conducted ablation experiments using five options. Each option involved gradually incorporating the functionalities of the modules in §IV and conducting topology discovery experiments. The fifth option included the complete integration of all the functionalities from the modules. Here are the specific implementation details for each option.

- **Option I**: Use the new addresses generated based on the existing prefix set in each round of probing without performing common prefix extraction and prefix expansion.
- **Option II**: Common prefix extraction is performed on the results of each round of probing, but no prefix expansion is performed. The target addresses for each round of probing are generated based on the set of common prefixes extracted in that round.
- **Option III**: Common prefix extraction and prefix expansion are performed in each round of probing, but no transition points are set. Further, a fixed value is used for the number of expansion bits, ranging from 1 to 4. The target addresses for each round of probing are generated based on the newly expanded prefix set.
- **Option IV**: Common prefix extraction and prefix expansion are performed on the probing results in each round. The expansion is based on fixed transition points with 8 predefined values (32, 36, 40, 44, 48, 52, 56, 60, 64).

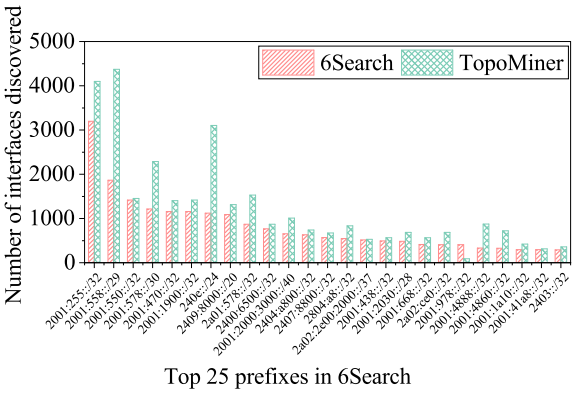

Fig. 8. Comparison of the number of interface addresses discovered by TopoMiner and 6Search under the top 25 prefixes with the highest number of addresses discovered by 6Search.

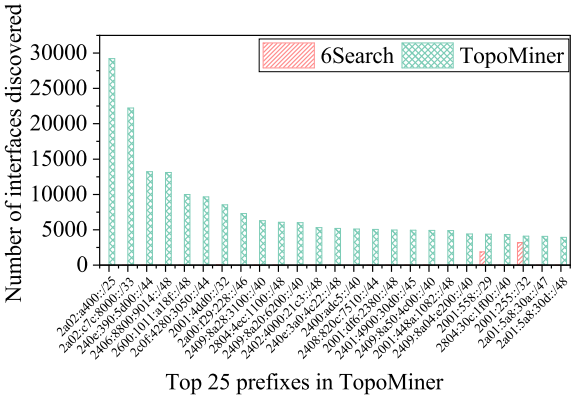

Fig. 9. Comparison of the number of interface addresses discovered by TopoMiner and 6Search under the top 25 prefixes with the highest number of addresses discovered by TopoMiner.

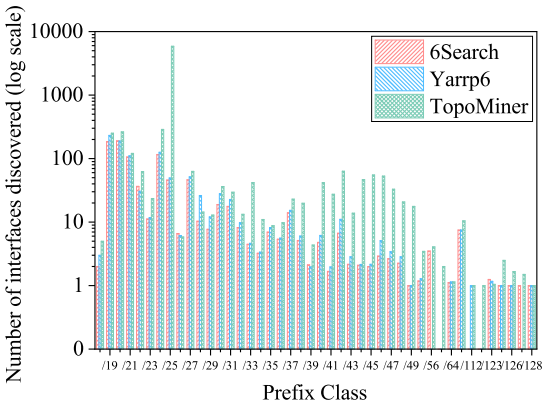

Fig. 10. Comparison of the number of discovered addresses under different matched categories of BGP routing prefixes.

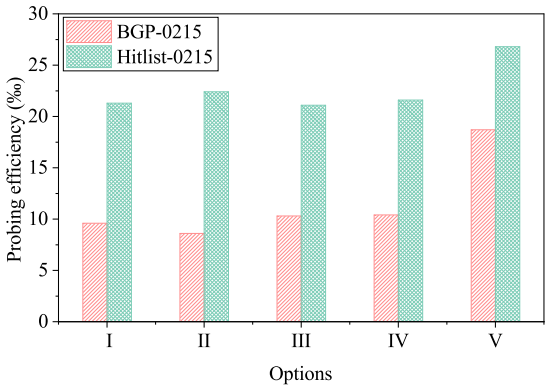

Fig. 11. Comparison of the probing efficiency of multiple options.

The target address for each round of probing is generated based on the newly generated prefix set.

- **Option V**: In each round of probing, common prefix extraction, prefix expansion, adaptive transition points, and expansion bits are combined. The target address in each round is generated based on the newly generated prefix set. Option V is the complete scheme of TopoMiner.

We conduct experiments on each of the above five options using two datasets, BGP-0215 and Hitlist-0215. The probing budget was uniformly set to 10M. Multiple experiments are conducted for each option, and the average probing efficiency is compared. The results, shown in Fig. 11, indicate that option V outperforms the other options in improving probing efficiency. This also implies that each design of TopoMiner contributes significantly. In addition, when considered in conjunction with Fig. 11, Fig. 6, and Fig. 7, TopoMiner's performance on the Hitlist-0215 dataset is significantly better than its performance on BGP-0215. This is intuitive as the Hitlist-0215 dataset provides more detailed information about address prefixes, making probing more efficient.

D. Ethical Considerations

Conducting large-scale, network-wide active measurements necessitates careful consideration of experimental methods to prevent causing harm. Therefore, we follow ethical conventions for network measurement, including recommendations provided by Partridge et al. [32] and Dittrich et al. [33] to perform IPv6 topology discovery. First, the network where our vantage point is located granted permission for us to conduct the measurement study outlined in this paper. Second, we chose to send ICMPv6 probes instead of UDP and TCP. This decision was based on the fact that ICMPv6 probes are relatively harmless, and they generate the most responses compared to UDP and TCP. Third, we use pseudo-random IIDs for topology discovery during probing. These IIDs are unlikely to correspond to active IPv6 hosts' addresses. The impact of using these pseudo-random IIDs on topology discovery is negligible compared to using ::1 IIDs.

VI. Conclusion

In this paper, we propose TopoMiner, an efficient topology discovery method specifically designed for IPv6 networks. TopoMiner is capable of conducting multiple rounds of probing using a given prefix set. It leverages the results ob-

tained from each probing round to identify regions with high-density interface addresses, allowing it to automatically adjust the breadth and depth of topology discovery. This approach enables a more comprehensive and in-depth exploration of network topology. Through real experiments, we demonstrate that TopoMiner outperforms state-of-the-art topology discovery methods, delivering superior results. Our experiments were conducted on a single server. In the future, we aim to improve our topology discovery by using multiple vantage points with higher privileges. This expanded approach will provide us with more comprehensive and accurate information about network topologies.

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